

Review the NREL white paper and assessment how these new approaches for “Climate Tech” or “Deep Tech” innovations are different from traditional methods such as the Diffusion of Innovation (DOI) Theory developed in 1962 by E. M. Rogers

The NREL white paper (NREL/MP-6A65-73438, April 2019) illustrates that hardware-focused “Climate Tech” or “Deep Tech” innovation faces unique challenges that set it apart from the classic social-technology adoption models described by E.M. Rogers’ Diffusion of Innovation (DOI) Theory.

Rogers’ DOI Theory maps the spread of innovations using five categories: innovators, early adopters, early majority, late majority, and laggards. It suggests that, after innovators, new technologies flow through these social groups, adopted first by risk-takers and influencers, then steadily by the mass population, and finally by those reluctant to change. This model assumes that once technology demonstrates clear advantages and reaches early enthusiasts, social influence and word-of-mouth drive it across the bell curve to widespread adoption.

However, deep tech and climate tech hardware do not follow this path easily. The NREL white paper explains that energy and climate technologies are often capital-intensive, technically complex, and require lengthy development and regulatory cycles before the public or traditional “early adopters” can engage. Challenges include the “Valleys of Death” where promising inventions cannot bridge the financial, technical, and demonstration gaps needed to reach the market.

To address these barriers, NREL and its partners have instituted multi-layered, proactive support mechanisms not accounted for in the original DOI framework. These include access to world-class laboratory resources, expert technical mentors, non-dilutive grant funding, early industry partnerships, and validation through demonstration projects. Programs like the Wells Fargo Innovation Incubator (IN2), Cyclotron Road, and Shell GameChanger Accelerator are designed specifically to help startups de-risk their technology, secure pilot deployments, and attract follow-on investment before mainstream adoption is possible.

Unlike the DOI approach where social momentum and peer influence are the main drivers, the novel NREL strategies focus on direct intervention and ecosystem building. These efforts actively align stakeholders from government, private industry, academia, and philanthropy to help startups overcome structural obstacles unique to physical and energy technology. Success in this model is measured not just by the willingness of early adopters, but by meeting technical milestones, attracting capital, and achieving real-world demonstration and policy approval.

In summary, while DOI theory is useful for explaining the adoption of low-risk consumer innovations, deep tech climate hardware requires much more deliberate, systematic, and collaborative support. The NREL model overcomes gaps that the DOI curve cannot bridge, offering a template for bringing critical climate technologies to scale in a way that is fundamentally different from traditional innovation diffusion.